FINAL PROJECT REPORT

SMART KHAREEDI

Smartphone Price Comparison Web Application

CCA 3 — Project Development, Presentation and Documentation

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Abstract

Smart Khareedi is a web-based smartphone price comparison application. It presents a collection of smartphone products and displays their stored prices from four retailer options: Amazon, Flipkart, Croma, and Vijay Sales. For each phone or variant, the application compares the prices available in its database and highlights the retailer with the lowest recorded price. A Buy Now link is provided for each retailer offer; selecting it opens the corresponding product page on that retailer's website when a valid product URL is stored. The project uses React.js for the frontend, Node.js and Express.js for the backend, and MySQL for data storage. Product details, prices, and product links are entered manually into the database. The current implementation does not use retailer APIs or automatically refresh prices.

1. Introduction

Smartphones are sold through multiple online retailers, and the price of the same model may differ between websites. Customers who want to find a suitable offer may need to visit several websites and compare prices manually. Smart Khareedi aims to simplify this task by collecting smartphone information and retailer offers in one interface. Users can view multiple phone options, compare the stored prices, identify the lowest recorded price, and open the relevant retailer product page.

2. Problem Statement

Checking several online stores separately to compare smartphone prices can take time and make it difficult to identify the lowest available recorded offer. Smart Khareedi addresses this problem by displaying retailer prices together and identifying the lowest price from the data maintained in its database.

3. Aim and Objectives

Aim: To develop a simple web application that compares stored smartphone prices across four retailer websites.

- Display multiple smartphone products on one website.
- Present prices from Amazon, Flipkart, Croma, and Vijay Sales where records are available.
- Compare prices for the same phone or variant using the database records.
- Highlight the retailer with the lowest recorded price.
- Provide a Buy Now link to the corresponding retailer product page.
- Maintain product details, prices, and links in a MySQL database.
- Provide a straightforward interface that can be opened directly without login or registration.

4. Project Scope

The current version includes the following functionality:

- Direct access to the main website without user login or registration.
- Display of multiple smartphone products.
- Four retailer options: Amazon, Flipkart, Croma, and Vijay Sales.
- Display of the stored price beside each retailer name.
- Lowest-price comparison and identification of the retailer with the lowest recorded price.
- Buy Now links to retailer product pages, based on URLs stored in the database.
- Manual maintenance of smartphone product information, prices, and links.

Out of scope in the current version: user accounts, login and registration, automatic price tracking, retailer API integration, and automatic database updates.

5. Technology Stack

Technology	Role in Smart Khareedi
React.js	Builds the frontend interface that displays phone listings, retailer prices, comparison results, and Buy Now links.
Node.js	Runs the backend JavaScript application.
Express.js	Defines API routes, handles search requests, queries MySQL, organizes results, and returns comparison data.
MySQL	Stores smartphone details, variants, retailer names, prices, availability, and product URLs.
MySQL Workbench	Used to manage and inspect the MySQL database.
HTML and CSS	Support page structure and visual styling within the web application.
Vite	Supports frontend development and running the React application locally.
Git and GitHub	Track code changes and support collaboration through a shared repository.
Visual Studio Code	Development environment used to edit and manage the project.

6. System Architecture

Smart Khareedi uses a frontend–backend–database architecture:

1. Frontend: React.js displays the smartphone catalogue, retailer options, prices, lowest-price information, and Buy Now links.
2. Backend: Node.js runs the server and Express.js handles API requests and comparison logic.
3. Database: MySQL stores the product records, variants, platform names, prices, and product URLs.

User → React.js Frontend → Express.js API (Node.js) → MySQL Database

When the application needs comparison information, the frontend requests data from the backend. The backend retrieves matching records from MySQL, organizes offers by product and variant, calculates the lowest eligible recorded price, and returns the results for display.

7. Main Features

Open access: The website opens directly and does not require users to create an account or sign in.

Smartphone catalogue: Multiple smartphone products are displayed on the website.

Four retailer options: Amazon, Flipkart, Croma, and Vijay Sales are used as retailer choices.

Price display: The stored price is displayed beside each retailer's name when an offer is present.

Lowest-price comparison: Prices available for the same phone variant are compared, and the lowest eligible recorded price is highlighted.

Buy Now navigation: The link opens the corresponding retailer product page when a valid product URL is stored.

Database-managed data: Product information, prices, and retailer links are manually entered and maintained in MySQL.

8. Database Design

The backend comparison query uses the following main database tables and fields. The table descriptions below reflect the schema referenced in the project's SQL query.

Table	Purpose
products	Stores product-level details such as product name, brand, model, category, image URL, and description.
product_variants	Stores configurations associated with each product, such as variant name, RAM, storage, and colour.
platforms	Stores retailer names, including Amazon, Flipkart, Croma, and Vijay Sales.
product_prices	Stores price entries linked to variants and retailers, including price, product URL, availability, and update timestamp.

Relationship overview: a product may have multiple variants. Each variant may have price records associated with different retailer platforms. The backend joins the related tables to create the data used for comparison.

9. Application Workflow

4. The user opens the Smart Khareedi website directly.
5. The frontend displays the available smartphone products.
6. The application retrieves product and retailer-price information from the backend, which reads the MySQL database.
7. The backend groups retailer offers under the relevant phone and variant.
8. The available stored prices are compared for the same variant.
9. The lowest eligible recorded price is identified and highlighted.
10. The user reviews prices from Amazon, Flipkart, Croma, and Vijay Sales where available.
11. When the user selects Buy Now, the browser opens the stored product page for the selected retailer.

10. Price Comparison Method

The comparison is performed on the price records stored in MySQL. For each phone variant, the backend considers offers with valid positive numeric prices and excludes offers marked “Out of Stock” when determining the lowest eligible price. It calculates the minimum of the remaining prices and identifies the retailer associated with that price. If no eligible offer exists, the lowest price is returned as unavailable.

Because the product data and prices are entered manually, the displayed comparison represents the database values and not an automatic real-time price check. The accuracy of the result depends on maintaining correct product variants, prices, and links.

11. Actual Result

The completed application opens directly without a login or registration page and displays multiple smartphone products. For each phone, the interface presents the available retailer names—Amazon, Flipkart, Croma, and Vijay Sales—with their stored prices and Buy Now links. The backend compares the prices recorded for the same phone variant and highlights the retailer offering the lowest eligible recorded price. Selecting Buy Now opens the corresponding product page on the chosen retailer's website when the stored URL is valid.

Example of the expected display (illustrative values only, not a measured test result):

Retailer	Illustrative stored price	Comparison
Amazon	₹79,999	—
Flipkart	₹78,499	—
Croma	₹77,999	Lowest illustrative price
Vijay Sales	₹80,990	—

The example prices above are illustrative and should be replaced with values from the team's current database or a screenshot of the running application before final submission if the report is expected to document a specific tested phone.

12. Testing and Verification

The following table records the expected behaviour of the current application. Verify each item in the running application before marking it as passed.

Test case	Expected result	Status
Open website	Main page opens without login or registration.	Observed per project description
Display products	Multiple stored smartphones appear.	Observed per project description
Retailer prices	Available stored prices appear beside retailer names.	Observed per project description
Lowest price	Lowest eligible price for a variant is highlighted.	Observed per project description
Buy Now link	Opens the relevant retailer product page when the URL is valid.	Verify each retailer link
Database source	Comparison results come from MySQL records.	Confirmed by shared backend code
Price freshness	Prices match the current retailer website.	Manual maintenance; not automatically verified

13. Team Details

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G27	Bhosale Sejal	1272250103	Team Leader
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G27	Bangad Parth	1272250059	Team Member
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All four members are part of Group G27. The group should describe each member's individual contribution accurately during evaluation. Specific individual tasks are not assigned in this report because a verified division of work was not provided.

14. Limitations

- Product information, prices, and retailer links are entered and updated manually.
- The application does not use retailer APIs or automatic price-refresh services.
- Stored prices may become outdated when retailer prices change.
- A Buy Now link depends on the correctness and availability of the stored product URL.
- The current version does not include login, registration, or personalised user accounts.

15. Future Enhancements

- Add a permitted and reliable mechanism for refreshing retailer prices automatically.
- Display a clear last-updated date and time for each price.
- Add filters for brand, budget, RAM, storage, and colour.
- Add sorting by price and retailer availability.

- Add price history and price-drop notifications.
- Improve mobile responsiveness and accessibility.
- Add automated tests for the comparison logic and retailer links.

16. Conclusion

Smart Khareedi provides a single interface for viewing smartphone products and comparing manually maintained prices from Amazon, Flipkart, Croma, and Vijay Sales. React.js builds the frontend, Node.js and Express.js run the backend and API, and MySQL stores the product and price information. The application highlights the lowest eligible stored price and provides links to retailer product pages. The project meets its current goal of simplifying side-by-side price comparison, while automatic price updates remain a possible future enhancement.

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